A native-grid learned inverse for multi-resolution ocean emulation

Abstract

The SamudraMulti model implemented on `main` used Perceiver modules to map heterogeneous physical grids into and out of a shared latent processor. Although this design allowed variable regular output query grids, it performed poorly even on one-step, one-degree prediction: full-data normalized mean-squared error (MSE) was `0.29469`, compared with a quoted external Samudra v2 reference of `0.0236`. Controlled auto-encoding experiments localized most of the unexpected error to the representation heads rather than the convolutional processor. In particular, fixed physical patches caused resolution-dependent information loss, while the Perceiver IO decoder had a narrow and weakly anchored route from latent tokens to spatial output queries.

The revised multi-resolution model retains an encoder–processor–decoder factorization but changes its semantics. A jointly trained pointwise encoder–decoder pair forms a learned approximate inverse on the native input grid and is then frozen; deterministic, channel-masked coordinate resampling provides output-grid flexibility; and a shared processor advances the state autoregressively in latent space. Boundary forcing and grid geometry enter each physical transition through separate paths rather than contaminating the reconstructive state. A completed one/half-degree run achieves aggregate normalized MSEs of `0.03982`, `0.05408`, and `0.06595` at physical leads 1, 2, and 4, respectively, while preserving same-grid reconstruction MSE near `1e-3`. These results support the new inverse and temporal contract, although remaining velocity-spectrum and route-exposure limitations should be resolved before quarter-degree training.

For reproducibility, “main” in this discussion means `origin/main` commit `d689f92c`, inspected on 2026-07-24. The comparison is specifically against its `SamudraMulti` implementation and default model configuration; Samudra v2 appears only as an external accuracy reference.

Architectural comparison

The `main` implementation compressed each fixed $3^\circ \times 5^\circ$ physical patch to a single learned vector before applying the spatial processor. Consequently, increasing input resolution increased compression without increasing the latent spatial grid: a patch contained 15 one-degree cells, 60 half-degree cells, or 240 quarter-degree cells. The decoder then reconstructed every output pixel through windowed Perceiver IO attention. Output latitude and longitude formed spatial queries, and each query produced all prognostic channels at once. This made output shape formally flexible, but required attention to learn both the physical correspondence between source and destination cells and the channel transformation. Boundary variables were encoded together with prognostic state, and conventional autoregression decoded a physical prediction and encoded it again at the next step.

The new model separates information representation, dynamics, and rendering:

$$z_0 = E_s(x_t), \qquad z_m = z_{m-1} + \alpha \odot P(z_{m-1} + E_b(b_m) + G(\ell)), \qquad \hat{x}_{t+N} = R_{\ell_z \to \ell_o}(D(z_N), M).$$

Here E_s and D are learned pointwise channel maps, P is the nonlinear U-Net processor, E_b encodes one time-aligned boundary state per transition, and G supplies position and cell scale to the processor. The residual scale α is initialized to zero. The renderer R performs periodic-longitude, physical-coordinate bilinear interpolation and renormalizes it independently with each prognostic channel's wet mask M. The selected implementation is defined by the native-grid model configuration and the one/half-degree training configuration; the previous representation is specified by the linked `main` snapshot above.

Property	SamudraMulti on `main`	Revised multi-resolution model
State encoder	Perceiver compression per fixed physical patch	Learned 1×1 map at every native cell
Spatial latent grid	Fixed by patch extent	Equal to the input grid
Decoder transport	Learned Perceiver IO query routing	Deterministic physical-coordinate resampling
Output channels	Joint vector from each spatial query	Joint pointwise projection before channel-specific transport
Geometry	Added to encoder and decoder tokens; coordinates form decoder queries	Zero-initialized processor sidecar
Boundary forcing	Mixed into encoder input	Separate encoder called once per physical step
Autoregression	Decode and re-encode physical state	Encode once and carry latent state
Zero-step behavior	Production forward always applies P; no reconstruction API	Frozen learned inverse $D(E_s(x))$
Output resolution	Flexible through query count	Flexible through requested coordinate arrays

The revised encoder is not an identity or "dumb" front end: its 160-channel representation is learned jointly with the decoder. What is removed is premature spatial compression. Spatial mixing is delegated to the dynamical processor, while the decoder learns channel transformations and grid geometry specifies transport.

Experimental interpretation

In a 32-sample autoencoder factorial derived from the `main` architecture, the one-latent direct-encoder/Perceiver-decoder arm reached MSE `0.27930`; increasing it to main's 256 decoder latents remained poor at `0.27802`. Direct/direct reached `0.01208`, localizing the unexpected failure to the decoder. In a matched one-degree forecast proxy, replacing only that decoder with resampling plus a pointwise projection reduced MSE from `0.38174` to `0.05166` (86.5%). Widening attention values corrected a channel bottleneck but not unanchored routing; position-anchored attention remained slower and less accurate than coordinate resampling with a learned encoder.

Multi-resolution controls exposed two further failures. Coarse patch tokens discarded half-degree bandwidth before decoding; retaining one learned vector per native cell restored it. Resampling latent mixtures before projecting prognostic channels also failed to reproduce channel-specific coastal masks. Reversing those operations removed 78% of the excess half-to-one error on identical weights. Both resolutions now use common one-degree normalization statistics.

Fixed-one-step forecast training exposed two temporal problems: a soft reconstruction term did not prevent encoder/decoder drift, and the processor was not trained to associate repeated calls with successive physical leads. The new procedure freezes E_s, D, then trains the transition, boundary encoder, geometry sidecar, and residual scale at true depths $\{1, 2, 4\}$. Depth N consumes N ordered boundary states and targets $x_{t+N\,dt}$.

Results and implications

The completed full-interval one/half-degree experiment trained for 6,392 optimizer updates. Aggregate validation MSE was `0.03982` at lead 1, `0.05408` at lead 2, and `0.06595` at lead 4, corresponding to reductions of 62.9%, 69.5%, and 73.9% relative to lead-matched persistence. Every one-to-one, one-to-half, half-to-one, and half-to-half route beat its persistence baseline at every evaluated lead. The one-to-one lead-one MSE was `0.02642`, 91.0% below the completed `main`-architecture full-data baseline of `0.29469`; because the training contracts differ, this magnitude summarizes integrated progress rather than a single-factor effect. Same-grid reconstruction remained exactly `0.001110` at one degree and `0.001243` at half degree throughout processor training, directly confirming that improved forecasts were not obtained by sacrificing the learned inverse.

Boundary ablations support the intended physical interpretation. Setting the per-step forcing to zero increased aggregate error by 23.4%, 61.1%, and 124.7% at leads 1, 2, and 4; reversing its temporal order increased error by 9.8%, 19.2%, and 93.7%. The transition therefore uses both boundary values and ordering. Spatial diagnostics are also substantially healthier than in the old patch-compressed model: temperature, salinity, and sea-surface-height high-wavenumber power ratios remain between `0.938` and `1.025` on all routes.

These comparisons should nevertheless be interpreted mechanistically rather than as a single controlled leaderboard result. The historical full one-degree run and the new multi-resolution run differ in training schedule, temporal supervision, normalization, and representation, so their headline MSEs do not isolate one factor. The decoder-only factorial and matched proxy provide the stronger causal evidence for the rendering change; the full run validates the integrated model. The full promotion run used one training seed; its independent endpoint audit confirms checkpoint-evaluation reproducibility, not run-to-run uncertainty. Two limitations remain. Half-degree same-grid velocity power is underrepresented (`0.790/0.751` for zonal/meridional velocity), suggesting insufficient processor exposure or capacity. Coarse-to-fine velocity power is lower still (`0.288/0.149`), an expected coarse-source and deterministic-prolongation limit of this architecture; a learned super-resolution head was not tested. In addition, one-to-one lead-one MSE is 28.7% above the completed single-resolution reference, slightly outside the provisional 25% degradation gate.

The principal conclusion is therefore architectural but bounded. The new native-grid inverse resolves the dominant representation and decoder failures while preserving flexible rectilinear output resolutions and a zero-to-N processor contract. The next step should keep this inverse fixed, test processor exposure or modest capacity increases, and validate conservative restriction for large downsampling ratios. Quarter-degree training is not yet warranted. The detailed experiments, definitions, and reproduction commands are collected in the decoder root-cause report.